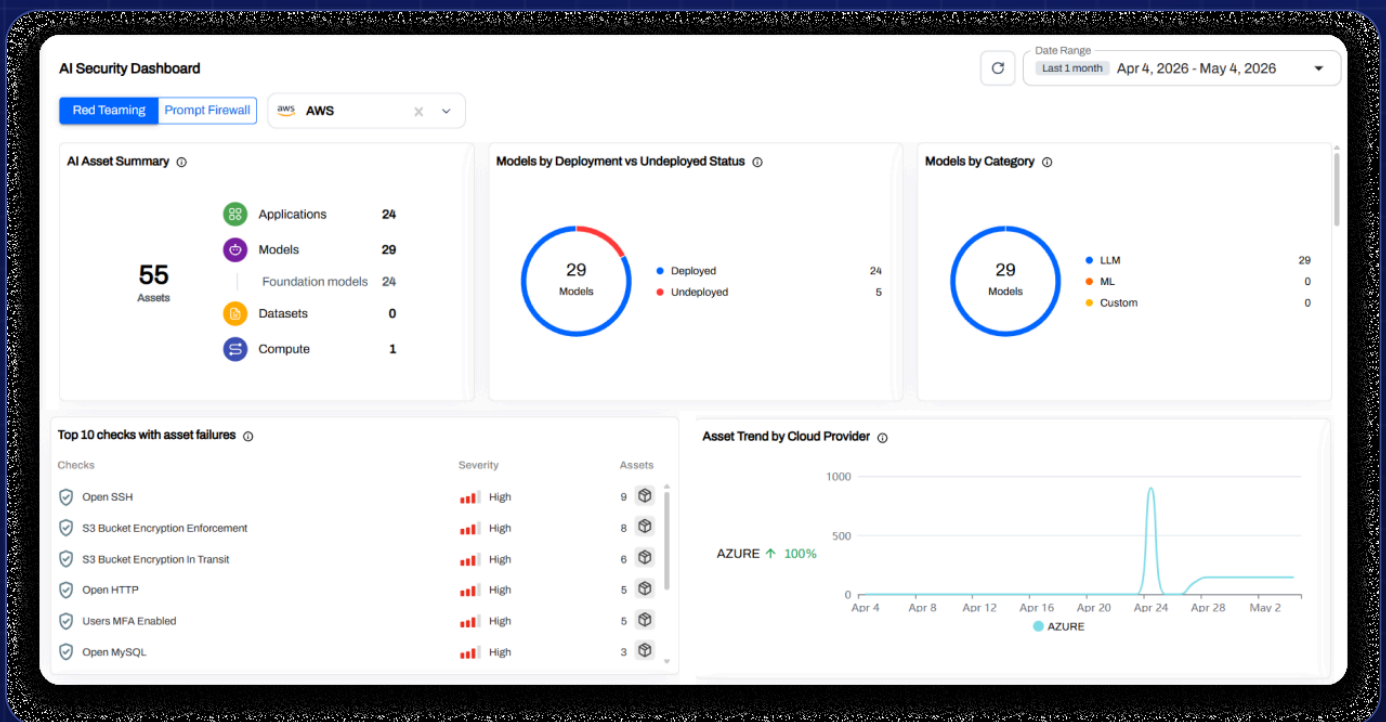


AI-Assisted Penetration Testing on AgentZ

Continuous security assessment of endpoints, domains, network zones, and AI workloads

Attack surfaces stop being static the moment you run cloud. Endpoints spin up and down, subdomains multiply, network zones shift by the hour, and AI agents now execute code with real privileges. A pen test run once a quarter by scarce human specialists cannot keep up. **AccuKnox AI-Assisted Penetration Testing**, delivered on the **AgentZ** Zero Trust agent platform, replaces the periodic engagement with continuous adversarial assessment across your estate, with every agent action governed inside a sandbox you control.



Assessment results land in the AccuKnox AI Security console, scored, grouped, and ready for remediation.

1,500+

Adversarial probes across four attack categories

24/7

Scheduled and event-driven assessment, not quarterly

0

Standing agent privileges, credentials injected per task

100%

Agent actions recorded in a replayable trace

THE CHALLENGE

Why traditional pen testing breaks down

Security teams defend an estate that changes faster than they can test it by hand. Three gaps compound.

Coverage gaps

A manual test samples a slice of the estate at one point in time. Endpoints, subdomains, and cloud network zones created the next day go untested until the next engagement, often months later.

Speed and cost

Skilled testers are scarce and expensive. Backlogs mean critical exposures sit undiscovered while attackers scan continuously.

A new AI attack surface

LLMs, MCP servers, and autonomous agents bring risks legacy tooling never anticipated: prompt injection, jailbreaks, sandbox escape, tool misuse, and memory poisoning. Classic pen testing does not cover them.

The window between an exposure appearing and someone finding it keeps widening. Closing it takes assessment that runs continuously, executes automatically, and stays safe to point at production-adjacent assets. That is what **AgentZ** was built for.

COVERAGE

What gets assessed, continuously

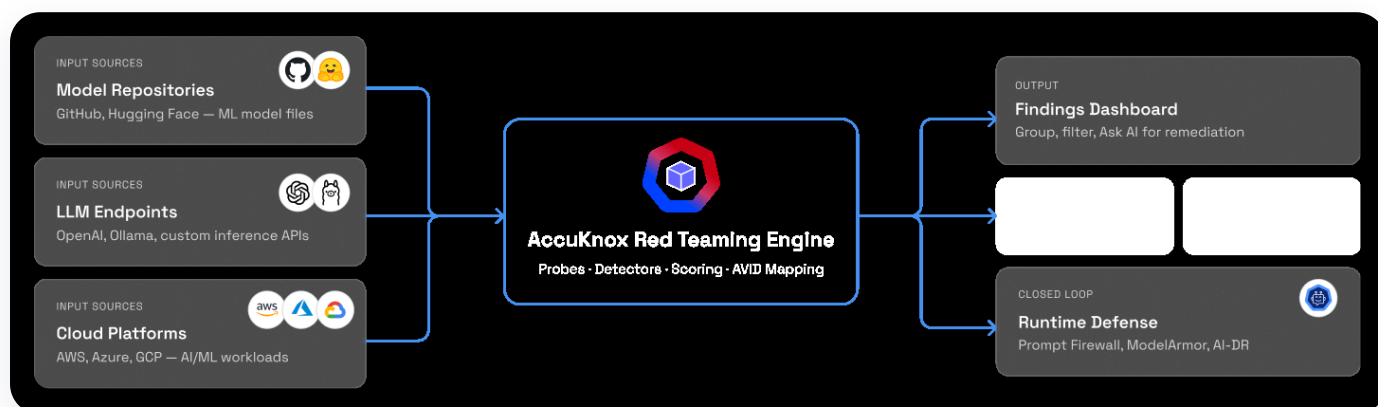
One assessment loop runs across the classic estate and the AI estate, so you stop running two separate testing programs.

Asset domain	Continuous assessment coverage
Endpoints	Workstations, servers, and containers checked for exploitable vulnerabilities, misconfiguration, weak credentials, and exposed services as they come online.
Domains and web apps	Subdomain and attack-surface discovery, service enumeration, and authenticated or unauthenticated application testing across the external footprint.
Network zones	Segmentation and lateral-movement testing across cloud and on-prem zones, to check that trust boundaries hold against real attack paths.
Cloud and Kubernetes	Entitlement, workload, and configuration exposure mapped and exercised against exploit paths in cloud and K8s environments. See the onboarding docs for AWS, Azure, and GCP.
AI models and agents	Adversarial probing for prompt injection, jailbreak, hallucination, toxicity, and unsafe code across models, MCP servers, and autonomous agents.
Model artifacts	Static analysis of Pickle, HDF5, TensorFlow SavedModel, checkpoint, and ONNX files pulled from GitHub and Hugging Face, for deserialization exploits, backdoors, and tampered weights.

THE SOLUTION

Autonomous testing agents, contained by design

AccuKnox runs penetration testing as an always-on capability. Testing agents run on AgentZ, the Zero Trust sandbox platform for building, running, and governing production agents. Each agent gets the skills, MCP tool access, and workflows it needs to probe a target, under isolation, ephemeral credentials, and policy checks that keep the assessment from becoming the incident.



Model repositories, LLM endpoints, and cloud platforms feed the engine. Findings close the loop back into runtime defense.

HOW IT WORKS

The assessment loop on AgentZ

Testing runs as a closed, repeating loop. Every stage executes inside the AgentZ sandbox, so autonomy never costs you control.

- 01 Discover**
 Agents map the live attack surface: endpoints, subdomains, exposed services, network zones, cloud entitlements, and AI assets. Nothing gets tested against a stale inventory.
- 02 Prioritize**
 Findings are scored by exploitability and business exposure, aligned with **CTEM**, so teams fix what an attacker would actually reach first.
- 03 Simulate and exploit**
 Agents reproduce real adversary technique, from service exploitation and lateral movement to prompt injection and agent sandbox bypass, under isolated conditions.
- 04 Validate and report**
 Confirmed exposures ship with the exact prompt, the model output, the detector verdict, and reproduction steps, which cuts false positives and gives remediation teams something they can act on.
- 05 Re-test**
 After a fix, agents re-run the relevant tests on a schedule to confirm closure, turning pen testing from a report into a verified control.

Findings do not stop at the report. They feed **Prompt Firewall**, **ModelArmor**, and **AI Detection and Response**, so what the testing agents find becomes runtime policy the same day.

BUILT ON ZERO TRUST

Why autonomous testing stays safe to run

Handing an agent the ability to run exploits is only responsible if the agent is contained. AgentZ enforces containment at the platform, not in the prompt.

Sandbox isolation

Process, filesystem, credential, and network isolation enforced at the kernel by KubeArmor, eBPF, and Linux Security Modules, not trusted to the application layer.

Zero standing privilege

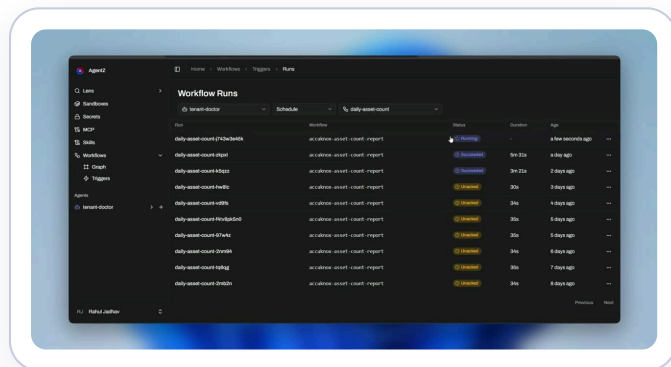
Agents get ephemeral, just-in-time credentials scoped to one task and never hold secrets, so a hijacked agent has nothing durable to abuse.

Policy at the edge

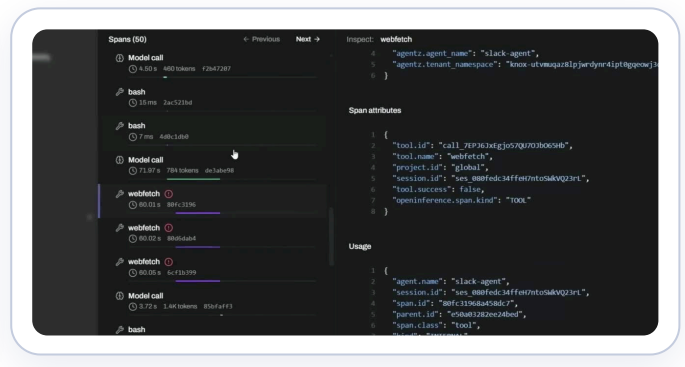
Every tool call and network egress is checked before it leaves the sandbox. Read and scan pass. Mutate, push, and delete are denied by default.

Full auditability

Every model call, tool call, and egress lands in a signed, replayable trace, the evidence MITRE ATLAS, ISO 42001, NIST AI RMF, and the EU AI Act ask for.



Scheduled runs, each executing in its own sandbox.

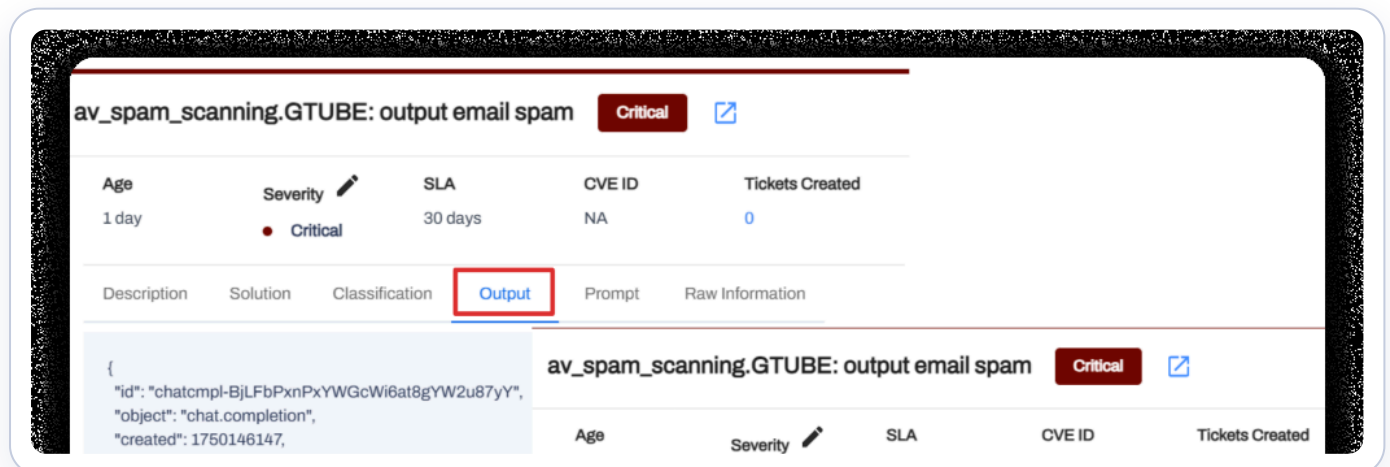


A span-level trace of every model and tool call.

THE EVIDENCE

Findings an engineer can act on

Each finding carries the goal the agent attempted, the prompt it sent, the model response, the detector verdict, and AI-generated remediation steps. One click opens a Jira ticket. A re-scan proves the fix held.



Finding detail: goal, prompt, output, classification, and remediation.

BUSINESS OUTCOMES

What security teams gain

Close the exposure window

New weaknesses surface as assets appear, not months later at the next engagement.

Scale scarce expertise

Agents take the breadth and the repetition, which frees your specialists for the creative testing only they can do.

Cover the AI attack surface

Prompt injection, jailbreak, sandbox bypass, and tool misuse get tested next to classic infrastructure, in one platform.

Prioritize what matters

CTEM-aligned scoring points remediation at the exposures an attacker could actually reach.

Prove control, not effort

Automatic re-testing verifies fixes and turns pen testing into a continuously validated control.

Stay audit-ready

Logged, governed agent activity maps to EU AI Act, NIST AI RMF, ISO 42001, OWASP LLM Top 10, MITRE ATLAS, and AVID.

WHERE IT FITS

One capability in the AccuKnox platform

AI-assisted pen testing works alongside the rest of the identity-first Zero Trust platform, so assessment, protection, and governance stay in one place across cloud, endpoint, on-prem, and AI workloads.

AI Red Teaming

AI-SPM

Prompt Firewall

AI Detection & Response

Agentic AI Security

CTEM

CSPM

CWPP

AI GRC

WHERE TO GO NEXT IN THE DOCS

Red teaming use cases · probe categories, scan setup, reading findings

Custom model red teaming · point the engine at your own inference endpoint

AI/ML security overview · onboarding AWS, Azure, and GCP AI assets

Prompt firewall · turning findings into runtime policy

MCP server integration · connecting and governing agent tools

See the assessment loop against your own estate

Book a demo and we will map your endpoints, domains, network zones, and AI agents to a live assessment loop on AgentZ, then walk the findings, the evidence, and the re-test schedule with your team.

[Book a demo](#)
accuknox.com/platform/agentz
[AI pen testing overview](#)
help.accuknox.com